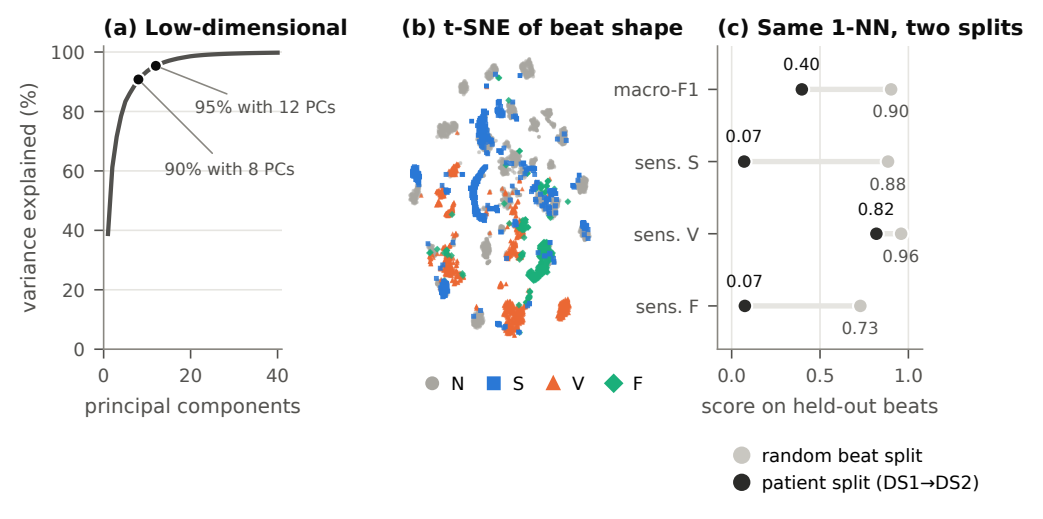

(a) Low-dimensional
variance explained (%)
100
80
60
40
20
0
95% with 12 PCs
90% with 8 PCs
0
20
40
principal components
(b) t-SNE of beat shape
N
S
V
F
(c) Same 1-NN, two splits
macro-F1
0.40
0.90
sens. S
0.07
0.88
sens. V
0.82
0.96
sens. F
0.07
0.73
0.0
0.5
1.0
score on held-out beats
random beat split
patient split (DS1→DS2)